

Spring 2025

Problem 1 (Sp25). Prove that there is no infinitely differentiable function on the real number line all of whose derivatives are 0 at 0 and positive everywhere else.

Problem 2 (Sp25). Let b_n be the fractional part of $(1 + \sqrt{3})^{2n+1}$. Show that $\lim_{n \rightarrow \infty} b_n = 0$.

Problem 3 (Sp06, Sp25). Evaluate the integral $\int_{-\infty}^{\infty} \frac{e^{itx}}{e^x + e^{-x}} dx$ for $t > 0$.

Problem 4 (Sp77, Sp25). Let a_n be defined by

$$1 - x^2 + x^4 - x^6 + \cdots = \sum_{n=0}^{\infty} a_n (x+2)^n \quad \text{for } 0 < x < 1$$

Find $\limsup_{n \rightarrow \infty} |a_n|^{1/n}$.

Problem 5 (Sp25). For a block matrix $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ of size $n \times n$ such that D^{-1} exists, prove that its determinant is equal to $\det(A - BD^{-1}C) \det D$.

Problem 6 (Sp25). Suppose that $\mathbb{F}$ is a field of characteristic p (0 or a prime). Show that there are $n \times n$ matrices A and B , with entries in $\mathbb{F}$, such that $AB - BA = I$ if and only if n is divisible by the characteristic of $\mathbb{F}$.

Problem 7 (Sp25). Prove that the conjugacy class in S_n of an even permutation σ *does not* remain a single conjugacy class in the alternating group A_n if and only if the cycle decomposition of σ consists entirely of distinct cycles of odd lengths.

Problem 8 (Sp25). How many maximal ideals are in the ring?

$$\mathbb{R}[x]/\langle x^{12} + 6x^2 + 5 \rangle$$

Problem 9 (Sp82, Fa08, Sp11, Sp25). For each of the following statements, either prove it or give a counterexample:

1. If $f_n(x)$ are continuous real valued functions on the unit interval, and f_n converges pointwise to the function f , that is, $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ for all x , then

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = \int_0^1 f(x) dx$$

2. If f_n are continuous and converges uniformly to a function f then the sequence $\int_0^1 f_n(x) dx$ has limit $\int_0^1 f(x) dx$.
3. If $g(m, n)$ is real for all integers m, n , and

$$\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} g(m, n) \quad \text{and} \quad \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} g(m, n)$$

are both defined, then they are equal.

4. If the real valued functions $h_n(x)$ are continuous on the unit interval, and $\lim_{n \rightarrow \infty} h_n(x) = h(x)$ for all x , then $h(x)$ is a continuous function of x .

Problem 10 (Sp25). Find the limit $\lim_{\varepsilon \rightarrow 0} \int_{-\infty}^{\infty} \frac{\varepsilon e^{-x^2}}{x^2 + \varepsilon^2} dx$ where the limit is taken for positive values of ε .

Problem 11 (Sp25). A complex function is holomorphic for $|z| > 2$ and in this region has derivative $e^z(z-a)/(z^2-z)$ where a is a complex number. Find a .

Problem 12 (Sp25). Suppose that $f(t)$ is an infinitely differentiable function of the real variable t , which vanishes for $t \geq 1$. Prove that

$$I(s) = \int_0^1 f(t)t^s dt$$

extends to a meromorphic function for all complex s , and find the possible values of s that can be poles.

Problem 13 (Sp25). Let $P(k)$ be a polynomial. Show that $S(n) = \sum_{k=0}^n P(k)$ is a polynomial.

Problem 14 (Sp25). Find the number of conjugacy classes of 4×4 complex matrices A with $A^4 = I$.

Problem 15 (Sp25). Find integers m, n with $0 < m < n$ such that $x^m \equiv x^n \pmod{2025}$ for all integers x . What is the smallest possible value of n ?

Problem 16 (Fa14, Sp25).

1. By counting the number of pairs (g, x) with $g \in G, x \in X$ and $g(x) = x$, show that the number of orbits of a finite group G acting on a finite set X is the average number of fixed points of elements of the group.
2. In how many ways, up to symmetries of the hexagon, can one color the vertices of a regular hexagon using 4 colors?
3. In how many ways, up to symmetries of the square, can one color the 4 corners of a square with 10 colors?